

Reproducing and Extending KELE: A Multi-Agent Framework for Structured Socratic Teaching with LLMs

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Abstract

We reproduce and extend KELE (Peng et al., 2025), a multi-agent framework for structured Socratic teaching with large language models. KELE decomposes Socratic dialogue into five teaching stages governed by SocRule, using a consultant–teacher architecture where one LLM selects teaching strategies and another generates responses. We reimplement the full pipeline, serve SocratTeachLLM locally via vLLM, and evaluate on the 681-dialogue test split of SocratDataset. Our primary reproduction with GPT-4o as live consultant achieves ROUGE-2 of 26.04 and BLEU-4 of 19.60. Working under a 32 GB single-GPU compute budget that forecloses the canonical two-model stack, we develop a *unified* architecture in which one open-weight backbone serves both roles through a single fusion structured-output call, eliminating per-run API spend and freeing VRAM for larger backbones. Across a smoke–mini–full progression spanning {Qwen3.6-35B-A3B, Qwen 27B Q5, Gemma 4 31B} $\times$ {think, no-think} $\times$ {two-call, fusion}, fusion-think uniformly dominates. Qwen3.6-35B-A3B fusion-think delivers a **+12.76-point absolute lift** in overall state accuracy on the full 681-dialogue test split (1.49 $\times$, with 3–5 $\times$ lifts on the harder middle and closure stages), and our headline candidate **Gemma 4 31B** surpasses A3B on every smoke and mini metric we measure, projecting a substantially higher full-run state accuracy at no additional cost. We also release **SocratDataset-EN**, a complete English translation of all 6,803 SocratDataset dialogues (see §3.2). Code, data, and trained checkpoints are publicly available.

1 Introduction

Dialogue plays a central role in shaping learning outcomes. Traditional knowledge-imparting instruction places students in a passive role, delivering answers directly without engaging deeper reasoning (Johnston, 1994). Socratic teaching offers an alternative: through structured questioning,

learners are guided to actively construct knowledge and discover answers themselves (Seeskin, 1987; Chang et al., 1998). Research consistently shows that Socratic dialogue promotes cognitive development and deeper understanding (Knezic et al., 2010), yet its reliance on highly skilled instructors makes it difficult to scale in practice.

Recent advances in large language models (LLMs) have opened a path toward automating Socratic instruction. LLMs are increasingly applied in educational settings—student Q&A, lesson planning, and adaptive tutoring (Kasneci et al., 2023; Gan et al., 2023)—and several systems have explored Socratic-style dialogue generation, including prompt-based approaches (Zhang et al., 2024; Al-Hossami et al., 2024) and fine-tuned models such as SocraticLM (Liu et al., 2024) and EduChat (Dan et al., 2023). However, these efforts generally focus on surface-level heuristic questioning while neglecting the structured, multi-stage progression that characterizes effective Socratic pedagogy.

Peng et al. (2025) address this gap with KELE (Knowledge-Enlightened Learning Enhanced by LLMs), a multi-agent framework that decomposes Socratic teaching into five hierarchical stages—student questioning, concept probing, inductive reasoning, rule construction, and teacher summary—governed by a formal rule system called SocRule containing 34 teaching strategies. The framework employs a “consultant–teacher” mechanism: a *consultant* agent evaluates student cognitive state and selects the appropriate teaching strategy, while a *teacher* agent generates the actual dialogue response. To power the teacher agent, the authors fine-tune GLM4-9B (Team GLM et al., 2024) using LoRA (Hu et al., 2022) on SocratDataset, a corpus of 6,803 dialogues with over 42,000 annotated turns, producing SocratTeachLLM—a model that outperforms GPT-4o across all nine Socratic teaching evaluation dimensions despite being significantly smaller.

In this work, we reproduce and extend the KELE framework with three goals:

1. **Reproduction.** We reimplement the consultant–teacher pipeline, serve SocratTeachLLM locally via vLLM (Kwon et al., 2023), and use GPT-4o (OpenAI, 2024) as the consultant to evaluate on the same 681-dialogue test split. We report ROUGE (Lin, 2004), BLEU (Papineni et al., 2002), and per-stage state accuracy to verify the original claims.
2. **Diagnosis.** We analyze where the reproduction matches or diverges from the published results, with particular attention to the low mid-stage state accuracy (stages c and d below 6%) that emerges from our evaluation.
3. **Improvement.** We propose replacing the LLM-based consultant with a lightweight supervised classifier trained on SocratDataset’s 42,000 state-labeled turns, targeting the system’s weakest link: the consultant’s ability to correctly identify student cognitive states during the complex middle stages of Socratic dialogue.

Table 1 summarizes our baseline reproduction against the original results. Our system uses the same configuration as Peng et al. (2025): GPT-4o as the consultant agent and SocratTeachLLM as the teacher agent, evaluated on the 681-dialogue test split (10% of SocratDataset). We include the paper’s GPT-4o teacher baseline for additional context.

2 Related Work

2.1 Large Language Models in Education

LLMs are rapidly being integrated across educational applications, from student Q&A and personalized tutoring to lesson planning and intelligent feedback (Kasneci et al., 2023; Gan et al., 2023). InstructGPT-style alignment through reinforcement learning from human feedback has enabled conversational agents that can adapt explanations to student needs (Chung et al., 2024). RetLLM-E (Mitra et al., 2024) retrieves relevant course material to ground LLM responses for student forum questions, while LessonPlanner (Fan et al., 2024) assists novice teachers in generating pedagogically sound lesson plans. The CLASS framework (Sonkar et al., 2023) builds intelligent tutoring systems grounded

System	R-1	R-2	R-L	BLEU-4
<i>Peng et al. (2025) — original results</i>				
GPT-4o (teacher)	48.25	22.55	38.27	29.93
SocratTeachLLM	57.40	33.63	50.77	41.96
<i>Our reproduction</i>				
SocratTeachLLM	44.61	26.04	38.02	19.60
<i>Delta (ours vs. original SocratTeachLLM)</i>				
	−12.79	−7.59	−12.75	−22.36

Table 1: Reproduction comparison on the 681-dialogue test split. R-1/2/L = ROUGE-1/2/L (F1). Both the original and our system use GPT-4o as the consultant and SocratTeachLLM as the teacher. Our ROUGE-L closely matches the paper’s GPT-4o teacher baseline (38.02 vs. 38.27), and our ROUGE-2 exceeds it (26.04 vs. 22.55), confirming that SocratTeachLLM provides meaningful gains over a general-purpose teacher. The BLEU-4 gap likely stems from undocumented generation parameters in the original work.

in learning science principles, guiding students step-by-step through interactive dialogue. These systems demonstrate the growing capacity of LLMs to serve as educational tools, though most operate in a single-agent paradigm without explicit pedagogical structure.

2.2 Technology-Enhanced Socratic Teaching

The Socratic method—teaching through guided questioning rather than direct instruction—has a long history in educational research (Seeskin, 1987; Chang et al., 1998). Technology-enhanced implementations range from early rule-based dialogue systems (Graesser et al., 2004) to modern LLM-powered approaches. Al-Hossami et al. (2024) demonstrate that LLMs can employ Socratic questioning to help students debug code, while Shridhar et al. (2022) show automatic generation of Socratic sub-questions for math word problems. The SPL system (Zhang et al., 2024) leverages GPT-4 for Socratic dialogue in a playground-style interface.

On the fine-tuning side, EduChat (Dan et al., 2023) and SocraticLM (Liu et al., 2024) train models on Socratic-style datasets to internalize heuristic questioning patterns, while Ding et al. (2024) apply the Socratic method specifically to conversational mathematics teaching. MathDial (Macina et al., 2023) contributes a tutoring dialogue dataset grounded in math reasoning with rich pedagogical annotations.

However, as Peng et al. (2025) observe, these approaches primarily focus on generating heuristic questions at the individual turn level without en-

forcing coherent multi-stage teaching progression. The absence of structured rules can lead to arbitrary guidance, abrupt topic shifts, and failure to consolidate learning. KELE addresses this through its five-stage SocRule system and dual-agent architecture, making it—to our knowledge—the first framework to systematically implement structured Socratic teaching with LLMs.

2.3 Multi-Agent Collaborative Systems

Multi-agent architectures have emerged as an effective paradigm for complex LLM tasks that benefit from separation of concerns (Ge et al., 2023). In the educational domain, the key insight of KELE’s consultant–teacher design is that teaching *planning* (what stage and strategy to use) and teaching *execution* (generating the actual dialogue) are fundamentally different capabilities that benefit from distinct agents (Peng et al., 2025). This mirrors real-world educational practice where teaching consultants advise on instructional strategy while classroom teachers handle direct student interaction (Nevin et al., 2009). Single-agent approaches struggle with both responsibilities simultaneously, often producing responses that are individually reasonable but lack coherent pedagogical progression (An et al., 2024). Our work investigates whether the consultant role—essentially a state classification task—can be more effectively handled by a specialized classifier rather than a general-purpose LLM.

3 Dataset and Methodology

3.1 SocratDataset

SocratDataset (Peng et al., 2025) comprises 6,803 multi-turn Chinese dialogues totalling 42,892 annotated interaction turns, drawn from elementary school science education. Each dialogue simulates a Socratic tutoring session in which a teacher guides a student from an initial question to a correct, self-discovered answer. Every turn is annotated with one of 35 pedagogical states organized across five progressive teaching stages (Table 2). Dialogues average 6.3 turns (range 5–12). The dataset is split 90/10 into 6,122 training dialogues and 681 test dialogues (seed = 42), following the original paper’s partition.

3.2 SocratDataset-EN: English Translation

As a secondary contribution, we produced **SocratDataset-EN**, a complete English translation of all 6,803 SocratDataset dialogues (42,892 turns).

SocratDataset is natively Chinese; no English version existed, which limits access to the corpus for researchers outside the Chinese NLP community and constrains the fine-tuning tooling available for a teacher base-model upgrade.

The translation was generated by a local Qwen3.5-9B instance served via llama.cpp, with temperature = 0.1 and thinking disabled for near-deterministic output. A two-tier checkpointing system (local every 5 records, HuggingFace every 50 records) enabled the pipeline to resume overnight runs without data loss. Per-record validation enforced JSON structure, turn-count matching, required-field presence, and zero Chinese characters in the translated fields. Nine records that exhausted all automated retries were manually translated and are flagged in the dataset metadata via a distinct `translation_meta.model` tag. Throughput was approximately 370–390 records per hour on an AMD Radeon R9700 (Q4 quantization). The dataset is publicly available at <https://huggingface.co/datasets/ulises-c/SocratDataset-EN>.

3.3 SocRule: Five-Stage Teaching Framework

KELE operationalizes Socratic pedagogy through SocRule, a rule system derived from the teaching science literature (Trowbridge et al., 2011) that decomposes each dialogue into five strictly ordered stages:

- (a) **Student Questioning** (2 states). The student poses an initial question; the consultant determines whether the question is clear enough to proceed.
- (b) **Concept Probing** (≤ 3 turns, 6 states). The teacher probes the student’s prior knowledge and identifies misconceptions through diagnostic questions.
- (c) **Inductive Reasoning** (≤ 5 turns, 22 states). The most complex stage: the teacher guides the student to form hypotheses, test predictions against evidence, and reason inductively toward correct explanations.
- (d) **Rule Construction** (≤ 3 turns, 4 states). The student synthesizes observations into a general rule or definition, with the teacher providing corrective scaffolding as needed.
- (e) **Teacher Summary** (1 state). Once the student arrives at the correct answer, the teacher

Stage	Phase	States	Turns
a	Student Questioning	2	6,803
b	Concept Probing	6	7,480
c	Inductive Reasoning	22	14,445
d	Rule Construction	4	7,361
e	Teacher Summary	1	6,803
Total		35	42,892

Table 2: SocRule stage overview. *States* = number of distinct teaching states per stage. *Turns* = total annotated turns in SocratDataset. Stage c dominates with 22 states and 33.7% of all turns, reflecting the complexity of inductive reasoning in Socratic dialogue.

summarizes the key concepts and concludes the dialogue.

Stage progression is strictly monotonic—once the dialogue advances from stage s_i to s_{i+1} , it cannot regress. Each state maps to a concrete teaching strategy (e.g., “provide a counterexample,” “ask the student to re-examine prior learning”) that dictates the teacher’s pedagogical move. A maximum of 8 teaching rounds per dialogue is enforced; if the limit is reached before stage (e), the system forces a transition to the final rule-construction state (d33) to ensure graceful conclusion.

3.4 System Architecture

KELE employs a consultant–teacher dual-agent architecture in which the two roles operate in sequence on every turn. Given a student utterance u_t and the full dialogue history $H_{<t}$:

Consultant. The consultant receives $(H_{<t}, u_t)$ along with the complete SocRule specification encoded in its system prompt. It outputs a structured JSON response containing (1) an *evaluation*: a free-text analysis of the student’s current cognitive state, and (2) a *state code* (e.g., c12) identifying the applicable teaching state. The state code indexes into a deterministic state-to-action mapping that selects the corresponding teaching strategy.

Teacher. The teacher receives $(H_{<t}, u_t)$ together with the consultant’s evaluation and the action string for the selected state. It generates a natural-language response—typically a single Socratic question—that implements the prescribed strategy while maintaining an encouraging, age-appropriate tone.

Enforcement. Our implementation enforces SocRule constraints programmatically: if the consultant proposes a state from an earlier stage than

the current one, the system overrides the regression and retains the current stage. This phase-monotonicity guard ensures strict forward progression regardless of consultant errors.

3.5 Reproduction Setup

We reimplement the full KELE pipeline and evaluate on the 681-dialogue test split under the following configuration:

- **Teacher:** SocratTeachLLM (9.4B parameters; GLM4-9B (Team GLM et al., 2024) fine-tuned via LoRA (Hu et al., 2022) on SocratDataset’s training split), served locally via vLLM (Kwon et al., 2023) on an NVIDIA RTX 5090 (32 GB).
- **Consultant:** GPT-4o-2024-11-20 (OpenAI, 2024) via the OpenAI API, matching the original paper’s consultant choice.
- **Inference:** Each test dialogue is run end-to-end through the live system. The consultant produces predictions from scratch on every turn—it does *not* receive ground-truth state labels.

This last point constitutes the most significant methodological difference from the original evaluation. Peng et al. (2025) feed ground-truth consultant labels directly to the teacher, measuring teacher generation quality in isolation. Our evaluation measures the full pipeline including consultant errors, producing a more realistic assessment of end-to-end system performance. As we show in Table 1, this accounts for the majority of the ROUGE gap between our results and the published figures.

3.6 Evaluation Metrics

We evaluate along two complementary axes:

Text overlap. We compute ROUGE-1, ROUGE-2, and ROUGE-L (Lin, 2004) using character-level tokenization (each Chinese character treated as a token), which is the standard convention for Chinese text evaluation. BLEU-4 (Papineni et al., 2002) is computed with word-level segmentation. These metrics measure how closely the generated teacher responses match the ground-truth responses in the dataset.

State accuracy. We compare the consultant’s predicted state code against the ground-truth annotation for each turn, reporting both overall accuracy

Consultant	R-1	R-2	R-L	B-4	St. acc	Cost
GPT-4o (ours)	44.61	26.04	38.02	19.60	25.94%	\$17.49
Qwen3.5:9B (V100)	43.72	24.87	36.76	18.63	18.93%	\$0
Qwen2.5:7B (M4 CPU)	43.57	24.90	36.91	18.56	15.16%	\$0

Table 3: Consultant sensitivity across the 681-dialogue test split. ROUGE/BLEU varies by at most 1.0 R-1 across all runs, while state accuracy drops 11 points as consultant capability decreases from GPT-4o to Qwen2.5:7B. Text-overlap is driven by the fixed SocratTeachLLM teacher; state accuracy is the consultant-sensitive signal.

and per-stage breakdowns. State accuracy is a critical diagnostic metric: an incorrect state prediction causes the teacher to receive the wrong teaching strategy, which can produce pedagogically incoherent responses even when the teacher model itself is capable. As we report in Section 4, accuracy varies dramatically across stages—from 95% on the binary stage (a) to below 6% on the 22-way classification in stage (c)—motivating our proposed improvement of replacing the LLM consultant with a supervised classifier.

3.7 Consultant Sensitivity Analysis

The \$17.49 cost per evaluation run limits iteration speed. To quantify whether a weaker, locally hosted consultant could substitute, we ran two additional full evaluations: Qwen3.5-9B served via vLLM on WAVE HPC (NVIDIA V100, 32 GB), and Qwen2.5:7B served via Ollama on an Apple M4 Mac Mini (CPU inference), with teacher inference on an AMD Radeon R9700 (32 GB, ROCm). Table 3 summarizes all three runs.

The central finding is that ROUGE and BLEU are *insensitive* to consultant quality: a free local 7B model costs only -1.0 ROUGE-1 and -1.0 BLEU-4 relative to GPT-4o. State accuracy, by contrast, degrades sharply with consultant capability, particularly on stage c (22 fine-grained inductive-reasoning states), which drops from 4.70% (GPT-4o) to 0.63% (Qwen2.5:7B). This confirms that the consultant bottleneck is a structured classification problem poorly suited to zero-shot prompting of general-purpose LLMs.

3.8 Proposed Improvements

We target three complementary weaknesses identified in the baseline evaluation.

Hierarchical state classifier. We propose replacing the LLM consultant with a supervised two-stage classifier built on a pretrained Chinese BERT

encoder (Devlin et al., 2019). A first classification head predicts the teaching stage (5-way softmax); a second head predicts the within-stage state conditioned on the predicted stage (up to 22-way for stage c). Both heads are trained jointly on the 42,000 labeled turns of the SocratDataset training split. A 125M-parameter classifier trained directly on in-domain labels should substantially outperform zero-shot LLM prompting on this structured prediction task while eliminating per-run API cost entirely.

Teacher base model upgrade. SocratTeachLLM is fine-tuned from GLM4-9B (C-Eval ≈ 81.5), which lags behind more recent Chinese-capable open-weight models. We propose re-running the SocratDataset LoRA fine-tune starting from deepseek-ai/DeepSeek-R1-Distill-Qwen-14B (C-Eval 91.8, MIT license), which inherits chain-of-thought reasoning behavior well-suited to multi-step Socratic dialogue. We will ablate against Qwen/Qwen3-14B (dense architecture, Apache 2.0) and the direct architectural successor of the current base, zai-org/GLM-4-9B-0414, to isolate the contribution of base-model capability from fine-tuning data quality.

Retrieval-augmented teacher. To address the surface-form gap that drives the BLEU-4 shortfall, we propose indexing the SocratDataset training split by topic and concept using a Chinese sentence encoder (BAAI/bge-large-zh-v1.5) and prepending the top-3 most similar past dialogues as few-shot exemplars at inference time. Retrieval injects corpus-specific phrasing style without additional fine-tuning and is composable with both the classifier and the model-upgrade improvements.

4 Evaluation and Results

This section reports the full experimental campaign behind our extension. We motivate the design choices forced by our compute budget (§4.1), describe the smoke-mini-full gating protocol (§4.2), report the smoke matrix that selected our promotion candidate (§4.3), the mini gate that validated it (§4.4), the headline full run (§4.5), and the Gemma 4 31B pivot that supersedes it as our forward candidate (§4.6). §4.8 summarizes findings and §4.9 the planned next runs.

4.1 Compute budget and architectural pivot

All open-weight runs in this section execute on a single NVIDIA RTX 5090 (32 GB VRAM)

shared between consultant and teacher. The canonical KELE stack co-locates SocraTeachLLM (≈ 19 GB) with a frontier consultant; running this with even a 14B consultant exceeds the VRAM budget once KV cache and structured-output decode buffers are accounted for. This constraint precludes one of our originally proposed improvement axes (re-running KELE with a stronger Chinese-capable teacher), and it forces a single-backbone design choice: one open-weight model must serve both roles.

We exploit this constraint as a research opportunity. We introduce a **fusion** (unified-call) variant in which the consultant’s structured state prediction and the teacher’s natural-language response are produced by a *single* JSON-schema-constrained LLM call. The schema requires fields {evaluation, state_code, teacher_response} and is enforced server-side via llama.cpp’s strict json_schema grammar. Fusion has two practical benefits in our setting: (i) it halves the per-turn round-trip and KV-cache duplication relative to two-call inference, and (ii) it enables the consultant to condition its state choice on the same chain-of-thought used to generate the teacher response, which we hypothesized would improve mid-stage classification.

4.2 Smoke–mini–full gating protocol

Each candidate configuration is gated through three escalating evaluation tiers before any full-test-set commitment:

- **Smoke** ($n=5$, ~ 33 turns, ~ 10 – 40 minutes): cheapest sanity check. Used to rank a wide configuration matrix.
- **Mini** ($n=25$, ~ 145 turns, ~ 30 – 60 minutes): stratified sample with state-distribution coverage. Used as the promotion gate to a full run.
- **Full** ($n=681$, $\sim 4,170$ turns, multi-hour overnight): canonical evaluation on the 681-dialogue held-out test split.

A central methodological observation from our campaign is that smoke and mini results bracket the eventual full-run number from opposite directions: the smoke is consistently optimistic (small- n favorable variance) and the mini is consistently pessimistic (stratified-sample worst-case). For our headline run (§4.5), the smoke–mini *average*

Model	Mode	R-1	R-2	R-L	B-4	St. acc	min
<i>Two-call architecture</i>							
Qwen 27B Q5	think	26.40	8.72	18.71	2.98	39.39	39.7
Qwen 27B Q5	no-think	29.99	12.36	21.91	5.22	39.39	38.6
Qwen 35B-A3B	think	27.52	9.93	19.57	4.29	24.24	10.6
Qwen 35B-A3B	no-think	28.36	10.48	20.12	4.33	31.25	9.3
<i>Fusion (unified-call) architecture</i>							
Qwen 27B Q5	think	31.88	14.08	23.76	6.09	46.88	21.0
Qwen 27B Q5	no-think	29.78	11.09	21.11	4.78	30.30	20.8
Qwen 35B-A3B	think	32.96	12.59	23.93	5.87	42.42	7.0
Qwen 35B-A3B	no-think	29.70	10.42	20.49	5.63	40.62	9.1

Table 4: Smoke matrix ($n=5$ dialogues, ~ 32 – 33 turns each). All cells use one open-weight model serving both consultant and teacher. “min” = wall-clock minutes. Fusion-think dominates two-call within each model, and Qwen 35B-A3B fusion-think wins on speed at competitive state accuracy. Qwen 27B fusion-think wins on smoke state accuracy (46.88) but at $3\times$ the per-turn cost of A3B due to dense (vs. A3B’s mixture-of-experts) decoding.

($+12.86$ vs. the realized $+12.76$) was the most accurate small- n predictor of full-run state accuracy. We adopt this average as our promotion criterion going forward.

4.3 Smoke matrix: identifying the best fusion configuration

Operating under the budget constraint of §4.1, we ran the full {Qwen 27B Q5, Qwen3.6 35B-A3B (Qwen Team, 2025)} $\times$ {two-call, fusion} $\times$ {think, no-think} matrix at smoke scale to identify the strongest configuration before committing GPU-hours to mini and full runs. All eight cells use the same model serving both consultant and teacher roles. Table 4 reports the eight smoke results.

Findings from the smoke matrix. (1) *Fusion uniformly improves state accuracy* over two-call within each model and thinking mode (mean $\Delta = +9.9$ points). The single-call architecture’s ability to condition state choice on the same reasoning trace used for response generation appears to matter more than the loss of an explicit consultant-then-teacher decomposition. (2) *Thinking mode helps within fusion*; it hurts within two-call. We attribute this to the fact that two-call’s consultant has its own dedicated thinking budget, so a second thinking pass on the teacher provides no marginal information; in fusion the single thinking trace is shared across both decisions. (3) *A3B’s MoE architecture pays off at scale*: 35B-A3B is $3\times$ faster per turn than 27B Q5 in fusion-think (7.0 vs. 21.0 minutes for $n=5$) because only $\sim 3B$ of its 35B parameters activate per token. We promote **A3B fusion-think** to the mini gate as the best speed–accuracy trade-

Stage	Baseline	A3B mini	Δ	Multiplier
a (probe)	95.15%	88.00%	-7.15	0.92×
b (early reasoning)	36.93%	32.14%	-4.79	0.87×
c (22-state induction)	4.70%	10.64%	+5.94	2.26×
d (resolution)	5.04%	13.04%	+8.00	2.59×
e (closure)	11.92%	54.55%	+42.63	4.58×
Overall	25.94%	35.17%	+9.23	1.36×

Table 5: Mini gate: per-stage state accuracy for Qwen 35B-A3B fusion-think ($n=25$, 145 turns) versus the GPT-4o baseline ($n=681$, 4,294 turns). Multi- $\times$ lifts on stages c, d, e — the harder pedagogical work — meet the promotion criterion.

off. The 27B fusion-think configuration is held in reserve as a high-accuracy fallback if A3B underperforms at full scale.

4.4 Mini gate: validating A3B for full-run promotion

The mini run for Qwen 35B-A3B fusion-think ($n=25$, 145 turns, 29 minutes 5 seconds) returned a state accuracy of 35.17% (Table 5), -7.25 points below the smoke and +9.23 points above the GPT-4o baseline. With smoke at +16.48 and mini at +9.23, the smoke-mini average was +12.86, which we registered as our predicted full-run lift before committing the overnight budget.

4.5 Full run: Qwen 35B-A3B fusion-think (16h 29m, $n=681$)

We launched the full evaluation at 19:46:53 PDT on 2026-05-04 and finished at 12:16:03 PDT the next day — **16 h 29 m 10 s** of contiguous compute, 4,171 evaluated turns, and 38 schema-fallback turns (0.91%, well under our 5% schema-reliability gate). Per-turn time averaged 14.2 seconds, +18% over the mini’s 12.0 s/turn. We retain $\times 1.18$ as the mini-to-full wall-clock margin for projection of subsequent overnight runs.

Table 6 reports the headline result against the GPT-4o baseline.

The ROUGE/BLEU trade-off. A3B’s pedagogically rich, paraphrastic Chinese diverges from the terse SocratTeachLLM-generated ground-truth phrasing in SocratDataset. We confirmed this is a stylistic plateau and not a model-size issue: the 27B Q5 fusion smoke ROUGE-1 (31.88) is below A3B fusion’s smoke ROUGE-1 (32.96), so further scaling within the Qwen family does not close the surface-form gap. The state-accuracy axis, which measures whether the consultant routed the dialogue through the correct pedagogical strategy, cap-

Metric	GPT-4o baseline	A3B fusion-think	Δ
<i>State accuracy (overall)</i>			
Overall	25.94%	38.70%	+12.76 (1.49×)
<i>Per-stage state accuracy</i>			
a (probe)	95.15%	91.78%	-3.37
b (early reasoning)	36.93%	39.29%	+2.36
c (22-state induction)	4.70%	17.57%	+12.87 (3.74×)
d (resolution)	5.04%	14.78%	+9.74 (2.93×)
e (closure)	11.92%	56.83%	+44.91 (4.77×)
<i>Text overlap</i>			
ROUGE-1	44.61	30.63	-13.98
ROUGE-2	26.04	12.28	-13.76
ROUGE-L	38.02	22.37	-15.65
BLEU-4	19.60	5.86	-13.74

Table 6: Headline full-run result ($n=681$, 4,171 turns for A3B; 4,294 turns for baseline). A3B fusion-think delivers a +12.76-point absolute lift in overall state accuracy with 3–5 $\times$ multipliers on the harder middle and closure stages, while ceding ≈ 14 ROUGE-1 points to the SocratTeachLLM-trained baseline.

Metric	A3B smoke ($n=5$)	Gemma smoke ($n=5$)	A3B mini ($n=25$)	Gemma mini ($n=25$)
<i>State accuracy</i>				
Overall	42.42%	51.52%	35.17%	41.89%
Stage a	80.00	100.00	88.00	96.00
Stage b	50.00	50.00	32.14	46.43
Stage c	23.08	30.77	10.64	17.02
Stage d	20.00	40.00	13.04	20.00
Stage e	75.00	75.00	54.55	52.17
<i>Text overlap</i>				
ROUGE-1	32.96	33.84	30.51	30.11
ROUGE-2	12.59	14.82	11.69	11.50
ROUGE-L	23.93	25.74	21.30	21.48
BLEU-4	5.87	7.02	5.46	5.47
<i>Wall clock</i>				
Total (min)	7.0	12.3	29.1	54.6
s/turn	12.7	22.4	12.0	22.1

Table 7: Gemma 4 31B versus Qwen 35B-A3B at smoke and mini scales, fusion-think configuration. Gemma wins on every state-accuracy metric except stage e mini and matches or beats A3B on every text-overlap metric, at the cost of $\sim 1.85\times$ slower per-turn inference (Gemma is dense; A3B is MoE).

tures the actual pedagogical value of the system independently of phrasing.

4.6 The Gemma 4 31B pivot

After locking the A3B headline, we integrated Gemma 4 31B (Gemma Team, 2025) as a candidate teacher-consultant backbone with the same fusion-think configuration and ran it through smoke and mini gates on 2026-05-05. Gemma surpasses A3B on every smoke and mini metric we measured (Table 7).

Smoke-mini average projection. Applying the smoke-mini-average heuristic from §4.2, Gemma’s projected full-run state accuracy is $(51.52 + 41.89)/2 = 46.71\%$, a projected lift of +20.77 points over the GPT-4o baseline — +8.01 points beyond A3B’s realized full-run lift. This is a mean-

ingful margin even after derating for projection error: the same heuristic predicted A3B’s full-run lift to within 0.10 points, so a Gemma full run that lands within a few points of the projection would establish a new state-of-the-art consultant on this benchmark for fully open-weight, single-GPU systems.

Wall-clock projection. At 22.1 s/turn at mini scale and applying the $\times 1.18$ mini-to-full margin observed for A3B, a Gemma full run is projected at $4,171 \times 22.1 \times 1.18 \approx 108,800 \text{ s} \approx \mathbf{30.2 \text{ hours}}$, roughly $1.8\times$ A3B’s full-run wall-clock. This fits in a single weekend slot.

Why Gemma beats A3B at this scale. A3B activates only $\sim 3\text{B}$ of 35B parameters per token; this is what makes it $\sim 1.85\times$ faster but also caps its in-context reasoning depth on the 22-state stage c classification. Gemma 4 31B is a dense 31B model, paying full per-token compute and in exchange producing a sharper structured-state distribution, particularly visible on stage c (+6.4 points at mini) and stage d (+7.0 points at mini). The schema-fallback rate is 0/148 at mini for Gemma, identical to A3B’s 0/145.

4.7 Cross-architecture comparison: all evaluated runs

Table 8 consolidates every run we executed, grouped by architecture and ordered chronologically within group. This is the complete experimental record behind our analysis.

4.8 Takeaways

Open-weight, single-GPU systems can outperform GPT-4o on this benchmark. Our locked headline (§4.5) and our forward candidate (§4.6) both clear the GPT-4o consultant on overall state accuracy by double-digit margins, while running entirely on a single 32 GB consumer GPU at zero per-run API cost. The harder middle stages where general-purpose LLMs collapse (stage c at 4.70%, stage d at 5.04%) are precisely where fusion-think open-weight backbones produce the biggest multipliers ($3\text{--}5\times$).

Fusion-think is the right architectural primitive under VRAM pressure. The smoke matrix shows +9.9-point average state-accuracy improvement from two-call to fusion within each model and thinking mode. Fusion is also $\sim 2\times$ wall-clock faster than two-call by eliminating round-trip

and KV duplication, with negligible loss in schema reliability ($\leq 1\%$ fallback rate at full scale).

Smoke-mini averaging is a cheap and accurate full-run predictor. Across our two A3B data points (smoke +16.48, mini +9.23, full +12.76), the smoke-mini average (+12.86) tracked the realized full-run number to within 0.10 points. This is a methodological contribution worth disclosing: small- n optimism and stratified-sample pessimism cancel, and one should not treat either tier alone as canonical.

ROUGE/BLEU is a stylistic plateau for the open-weight family. Across Qwen 27B, Qwen A3B, and Gemma 4 31B, ROUGE-1 stays in the 30–34 band on smoke and converges near 30.5 at scale. SocratTeachLLM was *trained* on the ground-truth teacher phrasing in SocratDataset; no open-weight base model we tested reaches that band by zero-shot prompting alone. The pedagogical signal is in state accuracy.

Compute economics. The headline A3B full run cost 0 USD in API spend (vs. \$17.49 for the GPT-4o baseline) and ≈ 16.5 GPU-hours of local compute. The projected Gemma full run costs 0 USD and ≈ 30 GPU-hours.

4.9 Next steps

The 2026-05-14 deliverable is gated on three follow-on experiments, in priority order:

1. **Gemma 4 31B fusion-think full run** ($n=681$, $\approx 30 \text{ h overnight}$). If the run lands within ± 3 points of the projected 46.71%, it supersedes A3B as our headline result and becomes the basis for the paper’s empirical claims.
2. **Surface-form recovery on the locked A3B run.** Two cheap, composable interventions target the ≈ 14 -point ROUGE-1 gap: (a) teacher-style prompt engineering with 3-shot ground-truth exemplars instructing terse, single-question responses ($\sim 30 \text{ min smoke}$); (b) LoRA fine-tune of A3B on $\langle \text{dialogue history, teacher response} \rangle$ pairs from the SocratDataset training split ($\sim 30 \text{ min training}$, no extra inference VRAM). SocratDataset-EN (§3.2) additionally makes this fine-tuning path accessible with English-language tooling. If LoRA closes even half the gap while preserving state accuracy, the combined system

Run	Config	<i>n</i>	turns	R-1	R-2	R-L	B-4	St. acc	Wall clock
<i>KELE reproduction (two-model stack on RTX 5090 + API consultant)</i>									
Baseline	GPT-4o + SocratTeachLLM	681	4294	44.61	26.04	38.02	19.60	25.94%	4h 34m
<i>Consultant sensitivity (open-weight consultants, fixed SocratTeachLLM teacher)</i>									
V100 (WAVE)	Qwen3.5:9B + STL	681	4294	43.72	24.87	36.76	18.63	18.93%	—
R9700/Mac M4	Qwen2.5:7B + STL	681	4294	43.57	24.90	36.91	18.56	15.16%	—
<i>Unified single-backbone, two-call architecture (smoke matrix)</i>									
Qwen 27B think	two-call, think on	5	33	26.40	8.72	18.71	2.98	39.39%	39m 44s
Qwen 27B no-think	two-call, think off	5	33	29.99	12.36	21.91	5.22	39.39%	38m 33s
Qwen 35B-A3B think	two-call, think on	5	33	27.52	9.93	19.57	4.29	24.24%	10m 36s
Qwen 35B-A3B no-think	two-call, think off	5	32	28.36	10.48	20.12	4.33	31.25%	9m 19s
<i>Unified single-backbone, fusion architecture</i>									
Qwen 27B fusion think	fusion, think on	5	33	31.88	14.08	23.76	6.09	46.88%	20m 59s
Qwen 27B fusion no-think	fusion, think off	5	32	29.78	11.09	21.11	4.78	30.30%	20m 46s
A3B fusion think (smk)	fusion, think on	5	33	32.96	12.59	23.93	5.87	42.42%	7m 00s
A3B fusion no-think	fusion, think off	5	32	29.70	10.42	20.49	5.63	40.62%	9m 07s
A3B fusion think (mini)	fusion, think on	25	145	30.51	11.69	21.30	5.46	35.17%	29m 05s
A3B fusion think (full)	fusion, think on	681	4171	30.63	12.28	22.37	5.86	38.70%	16h 29m 10s
<i>Gemma 4 31B (forward candidate, fusion-think)</i>									
Gemma fusion think (smk)	fusion, think on	5	33	33.84	14.82	25.74	7.02	51.52%	12m 18s
Gemma fusion think (mini)	fusion, think on	25	148	30.11	11.50	21.48	5.47	41.89%	54m 33s
<i>Gemma fusion think (full)</i>	<i>projected</i>	<i>681</i>	<i>≈4170</i>	—	—	—	—	<i>≈46.71%</i>	<i>≈30h</i>

Table 8: Complete experimental record. “STL” = SocratTeachLLM (teacher). All open-weight smoke/mini/full rows use a single backbone serving both consultant and teacher roles. The dashed line in the consultant sensitivity rows reflects that those runs were submitted as off-rig batch jobs (WAVE Slurm; M4 CPU+R9700) and detailed wall-clock is omitted. The projected Gemma full row is the open work for the next deliverable (§4.9).

would be Pareto-superior to GPT-4o on every reported metric.

- Hierarchical state classifier ablation.** The proposed Chinese-BERT two-stage classifier (§3.7) remains a high-value experiment for stage c specifically. We will train it on the 42,000 labeled training-split turns and run it as a drop-in consultant against a frozen Gemma 4 31B teacher, isolating the structured-classification signal from the generation signal.

We do not plan to revisit the 27B Q5 backbone: its smoke fusion-think state accuracy (46.88%) is competitive with Gemma’s, but its $3\times$ per-turn cost makes a full run economically dominated by Gemma at the same accuracy band.

5 Conclusion

We reproduced the KELE consultant–teacher framework and extended it under a single-GPU compute budget that foreclosed the canonical two-model stack. By collapsing consultant and teacher into a single open-weight backbone served through a fusion structured-output call, we recovered the framework’s pedagogical signal while eliminating per-run API spend. Qwen 3.6 35B-A3B fusion-think delivered a +12.76-point absolute lift in overall state accuracy on the full 681-dialogue test split

($1.49\times$, with $3\text{--}5\times$ multipliers on the harder middle and closure stages), and our forward candidate Gemma 4 31B surpasses it on every smoke and mini metric we measured, projecting an additional +8-point lift at full scale. The remaining headline weakness — a ≈ 14 -point ROUGE-1 gap against a SocratTeachLLM teacher trained directly on the ground-truth corpus — is the focus of the next deliverable, with prompt-engineering and LoRA interventions queued against the locked A3B run.

Beyond the empirical numbers, the campaign yielded two methodological contributions worth carrying forward: (i) a smoke–mini–full gating protocol with smoke–mini averaging as a low-cost, high-accuracy full-run predictor, and (ii) the fusion architecture as a practical primitive for running KELE-style multi-agent pipelines on consumer-grade hardware.

Limitations

Single-GPU evaluation. All open-weight runs execute on one RTX 5090 (32 GB VRAM). This precludes the canonical KELE stack with SocratTeachLLM teacher and a separately served frontier consultant; we mitigate this by collapsing both roles into one backbone via the fusion architecture (§4.1), but the comparison to KELE’s two-model setup is therefore architectural-pivot-with-gain rather than direct.

Headline forward candidate is unconfirmed at full scale. The Gemma 4 31B headline projection (+20.77 points over GPT-4o baseline) rests on smoke ($n=5$) and mini ($n=25$) data. Although the smoke–mini-average heuristic predicted A3B’s full-run number to within 0.10 points, a single calibration data point is insufficient to bound the projection error; the full Gemma run (§4.9) is the gating experiment for the 2026-05-14 deliverable.

ROUGE/BLEU surface-form gap. Our open-weight teachers paraphrase pedagogically rich but stylistically distinct Chinese relative to SocratDataset’s ground-truth phrasing, which depresses surface-form metrics (≈ 14 points on ROUGE-1, BLEU-4). State accuracy is the more decision-relevant metric in this setting, but readers who weight reproducible phrasing highly should view the ROUGE/BLEU figures as bounding the system’s adequacy on imitation rather than instruction.

Schema fallbacks. The fusion architecture’s 0.91% schema-fallback rate at full scale (38/4,171 turns) is not zero. Fallback turns default to “stay in current state” and depress per-stage accuracy slightly. Tighter grammar constraints or a JSON-repair retry could close this further.

Single language. All experiments are in Chinese (the SocratDataset native language). Generalization to English or to other pedagogical domains is untested. SocratDataset-EN (§3.2) partially mitigates this by making the full corpus available in English, enabling English-language fine-tuning and evaluation in future work.

Ethics Statement

This work uses a publicly released educational dialogue dataset (SocratDataset) collected and licensed by Peng et al. (2025) for academic research. No personally identifiable information is processed, and no human subjects were involved in the experiments reported here.

The intended use of the system is automated Socratic-style scaffolding for elementary-school science questions, deployed as a tutoring aid under teacher supervision rather than as an autonomous instructor. We caution against treating LLM-generated pedagogical content as authoritative without expert review: open-weight teachers occasionally produce factually plausible but incorrect explanations (a known limitation of zero-shot

LLM tutoring), and the structured state classification we evaluate measures pedagogical strategy selection rather than factual correctness of the generated content. Deployment in any consequential educational setting should pair the system with ground-truth answer keys and human instructor review of dialogue transcripts.

We release all code, configuration, and trained checkpoints openly to enable reproduction, scrutiny, and downstream improvement of these results.

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